Lecture 8 - Continuity and Differentiability

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Announcements

1. Exam 1 is next Tuesday (10/06/2026) in class.
2. Exam 1 Review Session is next Monday (10/05/2026) from 6:10-8:10 PM in Vincent Hall 1.
3. All quizzes will be graded by Friday Night. If you want me to look at your homeworks, please email me and I'll grade them specifically.
4. Practice Exam Solutions will be released after class today.
5. Compiled lecture slides also released after class today.

Quiz Time

Quiz Time! Let's get into neat rows and spread out - only then will I pass out the quizzes. DO NOT START OR FLIP OVER THE QUIZ UNTIL I OFFICIALLY START THE QUIZ. If you have any questions about the rules for quizzes, refer to the slide in Lecture 2 or ask me directly. No cheating as per usual.

Continuity

Continuity Recap I

Recall our definitions of continuity for a function $f(x)$:

1. $f(x)$ is continuous at $x = a$ if:
2. $f(x)$ is continuous on (a, b) if:
3. $f(x)$ is continuous *everywhere* if:
4. $f(x)$ is continuous if:

Continuity Recap II

Also recall that: if $f(x)$ and $g(x)$ are continuous at $x = a$ and $h(x)$ is continuous at $f(a)$ then:

- $f(x) + g(x)$ is continuous at $x = a$
- $f(x) - g(x)$ is continuous at $x = a$
- $f(x) \cdot g(x)$ is continuous at $x = a$
- If $g(a) \neq 0$, then $\dfrac{f(x)}{g(x)}$ is continuous at $x = a$
- $h(f(x))$ is continuous at $x = a$.

Continuity Recap III

Finally we determined a large family of continuous functions:

- Constant functions $f_c(x) = c$
- Identity function $f(x) = x$
- Polynomials
  $p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$
- Root functions $f(x) = \sqrt[n]{x}$
- Rational functions $\dfrac{p(x)}{q(x)}$
- Trigonometric functions
  $\sin(x), \cos(x), \tan(x), \arcsin(x)$ etc.
- Exponential functions $f(x) = c^x$
- Logarithmic functions $f(x) = \log_c(x)$

Classifying Failures of Continuity: Jump Discontinuities

What does discontinuity look like?

Jump Discontinuity

A function $f(x)$ has a *jump discontinuity* at $x = a$, if the value of $f(x)$ jumps as we pass over $x = a$. Explicitly:

1. The function limits towards a value from the left: $\lim_{x \to a^-} f(x)$ exists
2. The function limits towards a value from the right: $\lim_{x \to a^+} f(x)$ exists
3. Those values do not agree:

$$\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x) \tag{1}$$

Classifying Failures of Continuity: Jump Discontinuities Visually

Visually, jump discontinuities look like:

Classifying Failures of Continuity: Removable Discontinuities

Best kind of discontinuity.

Removable Discontinuity

A function $f(x)$ has a *removable discontinuity* at $x = a$ if:

1. $\lim_{x \to a} f(x)$ exists
2. $f(a)$ either does not exist (we specifically call this a *hole discontinuity* or just a *hole*) or is not equal to $\lim_{x \to a} f(x)$.

Can redefine $f(x)$ *at a single point* to make it continuous at $x = a$.

Removable Discontinuities Visually

Visually,

Classifying Failures of Continuity: Infinite Discontinuities

Infinite Discontinuity

A function $f(x)$ has an *infinite discontinuity* if, equivalently:

- $f(x)$ has a vertical asymptote at $x = a$ or
- $\lim_{x \to a} |f(x)| = +\infty$

Infinite Discontinuities Visually

Visually,

Classifying Failures of Continuity: Essential Discontinuities

The *WORST* kinds of discontinuities.

Essential Discontinuity

A function $f(x)$ has an *essential discontinuity* at $x = a$ if equivalently

- At least one of the one-sided limits do not exist (not even diverging to $\pm infty$) or
- As $x \to a$, $f(x)$ takes on every number in some interval (v_{min}, v_{max}) *infinitely many times*

Essential Discontinuities Visually

Visually,

What does Continuity buy us?

We already saw one important use of continuous functions:

Direct Substitution

If $f(x)$ is continuous at $x = a$ then:

$$\lim_{x \to a} f(x) = f(a) \tag{2}$$

Continuous functions also have another very important property:

Intermediate Value Theorem

If $f(x)$ is continuous on (a, b) then (assuming $f(a) \leq f(b)$)
for any intermediate output c so that $f(a) \leq c \leq f(b)$, there must an input x_0 so that $a \leq x_0 \leq b$ and $f(x_0) = c$.

Tagline: all intermediate output values must be hit by intermediate input values.

IVT Example: Root finding I

Question: For the function

$$f(x) = x^3 - 3x^2 + 2x + 1, \tag{3}$$

which of the following intervals must contain a root of $f(x)$?

1. $[-2, -1]$
2. $[-1, 0]$
3. $[0, 1]$
4. $[1, 2]$

Approach 1: Solving. IDK how to factor general cubics!
Approach 2: IVT. We want to find an interval which contains a value x_0 so that $f(x_0) = 0$. Note: if $a < 0$ is negative and $b > 0$ is positive, then 0 is always inside $[a, b]$.

IVT Example: Root Finding II

Goal: find which of the intervals

1. $[-2, -1]$
2. $[-1, 0]$
3. $[0, 1]$
4. $[1, 2]$

makes

$$f(x) = x^3 - 3x^2 + 2x + 1 \tag{4}$$

switch signs at the endpoints.

1. $f(-2) = -8 - 3 \cdot 4 - 2 \cdot 2 + 1 = -8 - 12 - 4 + 1 = -23$
2. $f(-1) = -1 - 3 - 2 + 1 = -5$
3. $f(0) = 1$ ROOT!
4. $f(1) = 1 - 3 + 2 + 1 = 1$
5. $f(2) = 8 - 12 + 4 + 1 = 1$

Only know that a root must be inside $[-1, 0]$.

Differentiability

Differentiability Recap

Again, recall that:

1. $f(x)$ is differentiable at $x = a$:
2. $f(x)$ is differentiable on (a, b):
3. $f(x)$ is differentiable everywhere:
4. $f(x)$ is differentiable

Classifying Failures of Differentiability: Discontinuities

We can already classify some behavior making functions differentiable.

Recall: if $f(x)$ is not *continuous* at $x = a$ then it cannot be *differentiable* at $x = a$.

Classifying Failures of Differentiability: Corners

We can classify the rest of the behavior by examining the *continuity* of the derivative function $f'(x)$:

Corner Singularity

For a function $f(x)$ continuous at $x = a$, we say $f(x)$ has a *corner singularity* or *corner* at $x = a$ if the derivative function $f'(x)$ has a *jump discontinuity* at $x = a$.

Visually,

Classifying Failures of Differentiability: Cusps

Similarly,

Cuspidal Singularity

For a function $f(x)$ continuous at $x = a$, we say $f(x)$ has a *cuspidal singularity* or *cusp* at $x = a$ if the derivative function $f'(x)$ has an *infinite discontinuity* at $x = a$ *AND* does not diverge to $\pm\infty$. (e.g. the one sided limits separately go to $+\infty$ and $-\infty$).

Visually,

Classifying Failures of Differentiability: Vertical Tangents

Vertical Tangent Singularity

For a function $f(x)$ continuous at $x = a$, we say $f(x)$ has a *vertical tangent singularity* or a *vertical tangent* at $x = a$ if:

$$\lim_{x \to a} |f'(x)| = +\infty. \tag{5}$$

Visually,

Classifying Failures of Differentiability: Endpoint Singularity

Kind of a non-problem singularity:

Endpoint Singularity

For a function $f(x)$ defined on a *closed* interval $[a, b]$, we say that $f(x)$ has an *endpoint singularity* at the endpoints $x = a$ and $x = b$ just because two sided limits do not make sense at those points.

Visually,

Continuous Differentiability I

What about the last kind of discontinuity: removable discontinuities?
The last thing that could go wrong would be if $f'(x)$ had a hole discontinuity at $x = a$ - not often discussed as a class of singularity.
Instead, we care when everything goes *right*:

Continuous Differentiability

A function $f(x)$ is *continuously differentiable* at $x = a$ if $f'(x)$ is continuous at $x = a$.
Similarly, continuously differentiable on (a, b), , everywhere, etc.
A function $f(x)$ is *continuously differentiable* or *of class* C^1 if is is continuously differentiable at every point in its domain.

Intuituion: Functions whose graphs are continuous, and corner/cusp/vertical tangent free.

What does Continuous Differentiability buy us? I

Since $f(x)$ being continuously differentiable implies $f'(x)$ is continuous, we get the following by applying IVT to $f'(x)$:

Rolle's Theorem

For $f(x)$ continuously differentiable, and x_0 and x_1 two roots of $f(x)$ so that $x_0 \leq x_1$, there always exists an a such that $x_0 \leq a \leq x_1$ and $f'(a) = 0$.

What does Continuous Differentiability buy us? II

This also gives us a *VERY* important theorem:

Mean Value Theorem

For $f(x)$ continuously differentiable, and $(x_0, f(x_0)), (x_1, f(x_1))$ so that $x_0 < x_1$ there is always an a so that $x_0 \leq a \leq x_1$ and, equivalently:

- The tangent line $y = f'(a)(x - a) + f(a)$ is parallel to the secant line $y = \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0) + f(x_0)$ or
- $$\frac{f(x_1) - f(x_0)}{x_1 - x_0} = f'(a). \tag{6}$$

Uses of the Mean Value Theorem

Why care about MVT?

1. If $f'(x)$ is the zero function $f'(x) = 0$, then $f(x)$ must be constant!

2. Extremely useful for doing derivations involving derivatives: e.g. determining quadratic approximations!

3. Lots of the really useful tools in multivariable calculus need to use MVT to even work.

“Fun” application:

Worksheet Time

Let’s get into groups and work on the worksheet!

End of Class!

That’s all folks!